

Programme Q: selection after cancellation, analytic admissibility, and the first logarithmic energy correction

Claude, with Daniel Murfet

2026-09-09

Abstract

After Programme P the user deferred the geometric bridge (resolution of singularities) and asked for the rest of §3 to continue. Astra #38 supplied a bounded plan under a 24-unit cap: N1, the first nonzero admissible term after cancellation and superpolynomial flatness; N3, analytic families as admissible data; N2, the exact energy identity and the first logarithmic correction obtained by coefficient algebra rather than by differentiating a remainder; N4, positive normal boxes; N8, a reserve on zero-noise continuity. Fourteen theorem units (306–319) and four reviews (v37–v40, all PASS; v40 the release audit) later, the seabed holds the chart-level correction $N \log N (\mathcal{Z}_N[K \circ \pi]/\mathcal{Z}_N - \lambda/(\beta N)) \rightarrow -(m-1)/\beta$ for every multiplicity $m \geq 1$, the exact dilation law for boxes $(0, b]^d$, and the selection theorem for the first nonzero admissible pair.

1 What changed in the mathematics

The paper states $\mathbb{E}[K] = \lambda/n - (m-1)/(n \log n) + o(1/(n \log n))$ and obtains it by differentiating the asymptotic expansion of $\mathcal{Z}_n$ in n . That step is not justified in general, and review v38 forbade it. The formal route instead transports coefficients: the energy insertion is the exact weight shift $h \mapsto h+2k$ (an identity of integrands), the state-density representation commutes with the unit shift of its weights, and the Mellin moment recurrence $M(\mu+1, i) = (\mu M(\mu, i) - i M(\mu, i-1))/\beta$ turns into the coefficient recurrence $C_K(\mu+1, j) = (\mu C(\mu, j) - (j+1)C(\mu, j+1))/\beta$. With $C(\lambda, q) = 0$ above $m-1$ this gives $A_K = \lambda A/\beta$ and $B_K = (\lambda B - (m-1)A)/\beta$ for the two leading coefficients, and the quotient algebra $NL(\mathcal{Z}_K/\mathcal{Z} - \lambda/(\beta N)) = (A(B_K + \varepsilon_K) - A_K(B + \varepsilon))/(A(A + (B + \varepsilon)/L))$ on an eventual set where $\mathcal{Z} \neq 0$ closes the theorem. The remainder control needed at the next-log scale, and the log-weighted control needed for $m=1$, both come from the same positive power gap $\lambda + 1/Q > \lambda$ of the cutoff expansion; convergence at the leading scale alone would not have sufficed, and the exact energy identity $\mathcal{Z}_K = -(1/\beta)\mathcal{Z}'$ is proved as a theorem about integrals but is never used to differentiate an expansion.

Selection after cancellation (N1) is a finite combinatorial statement about the admissible index set $\Lambda \cap [0, L] \times \{0, \dots, D\}$ under the order “smaller exponent, then larger log degree”: a globally first nonzero pair exists uniquely whenever any coefficient is nonzero, and a cutoff expansion is asymptotically equivalent to that term; when every admissible coefficient vanishes the integral is $o(N^{-L})$ for every L , which does not make it zero (e^{-N} is flat and nonzero).

Analytic families (N3) close a hypothesis Programme P had left open: a family F_v holomorphic on a common polydisc of radius $R > 1$, jointly continuous and bounded on a smaller radius $1 < r < R$, defines continuous zero-noise data in the weighted- ℓ^1 topology (the geometric majorant $Mr^{-|r|}$ dominates the Cauchy coefficients uniformly), so the tangential integration and assembly theorems of Programme P apply with $\text{Re } F_v$ as the represented amplitude.

Positive boxes (N4) are an exact dilation $\mathcal{Z}_b(N; \eta) = b^{|h|+d} \mathcal{Z}_1(Nb^{2|k|}; \eta(b \cdot))$, so the leading coefficient on $(0, b]^d$ is $b^{|h|+d-2|k|\lambda} A(\eta(b \cdot))$ and the lower log coefficients are the binomial re-expansion of $\log N + 2|k| \log b$ already built into the box coefficient of Headline XXVIII.

2 Lessons

- *Transcribe interfaces from the file, not from memory.* The one blocking item in the release audit was a minus sign I typed into the frozen-interface block of the review prompt; the Lean definition has the positive shift. The reviewer could not see the repository and rightly held the release until the sign was reconciled. Copy definitions with `grep`.
- *The remainder scale is part of the statement.* Review v39 distinguished three scales (leading, next-log, log-weighted) and the proof needed the stronger two. Generalising `population_remainder_tendsto` to divide by L^j and to multiply by L^j cost one lemma each because the underlying estimate was a power gap; had it been a leading-scale limit only, the theorem would have been unprovable from it.
- *set shadows terms introduced later.* Abbreviations made with `set` before a `rw ... at` broke `div_self` and `mul_div_cancel_left` three times; fold with `simp only [← hdef]` or avoid `set` in algebraic closers.
- *Integral notation swallows subtraction.* `..., a - b * ...` parses the tail into the integrand; parenthesise the first integral.
- *Root-build name clashes are silent until the full build.* Two clashes (`clampCube`, `population_coeff_eq_zero_of_lt`) passed per-module builds; the exit-gated full build before every push caught both.
- *Do not restate a lemma under new names to spend budget.* The assembled energy correction was left out on the reviewer's advice: the missing pieces are the paired numerator/denominator assembled coefficient identification and the assembly bridges, not residual decay, and a wrapper would have proved nothing new.

3 Hand-off

Branch `tide/programme-q` merged into `main`; `HEADLINES.md` carries the Programme Q block with the release paragraph and fourteen non-claims; the paper mirror `grammar_lean.tex` receives dots for the selection theorem, flatness, analytic admissibility, the exact energy identity, the chart-level correction, the box dilation and the zero-noise continuity corollaries; the hand-off report is `projects/grammar/staging/programme-q-report.pdf`. Not undertaken: parity (N5), wall-crossing (N6), all-orders inverse-log division (N7, NO-GO), and the assembled next-log correction. The geometric bridge remains external by the user's instruction.